

NARESH KOTHARI

Automotive System Test Engineer

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PROFESSIONAL SUMMARY:

Detail-oriented Automotive Test Engineer with 3 years of experience in evaluating and validating ADAS and ECU functionalities. Skilled in requirement analysis, HIL testing, defect tracking, test automation using Python and CAPL, report generation. Proven ability to enhance system performance through rigorous testing and automation solutions.

WORK EXPERIENCE:

System Test Engineer | Magna Electronics (via Navtech Electronics)

Project 1: BMW ADCAM (Autonomous Driving Camera Module)

- Performed system-level testing and validation of ADAS functionalities using closed-loop and open-loop methodologies.
- Evaluated lateral control features such as Lane Keep Assist (LKA), Lane Change Warning (LCW), and Lane Support Assist (LSA) through simulated driving scenarios.
- Conducted HIL simulations by replaying real-world driving videos using the vehicle environment in front of the ECU camera.
- Developed and executed test cases based on system requirements and analyzed test results.
- Hands-on experience with Vector CANoe, including trace analysis, graphic panels, measurement setup, and test configuration.

Project 2: MAZDA SVS – Surround View System

- Performed system testing of ECU software functions such as Dirty Lens Detection (DLD), Pedestrian Detection (PD), Parking Line Detection (PLD), Failsafe, Calibration, HMI and Overlays.
- Developed and maintained detailed test cases and scripts for various modules based on system requirements.
- Conducted KPI testing for PLD by manually playing recorded videos in ADTF to verify:
 - Lighting and weather conditions
 - Distance from vehicle center to parking lines
 - Line width, visibility accuracy, and detection stability
- Performed load testing using the RRS setup to evaluate ECU performance.
- Assisted in automating regression test scenarios using CAPL scripting in CANoe.
- Developed Python-based automation scripts to execute HMI and failsafe test cases via CANoe, capture images, and generate consolidated PDF test reports.
- Executed automated test suites using pytest and used Jenkins for scheduled automation and reporting.
- Logged, tracked, and reported defects using Windchill, ensuring timely resolution and quality compliance.
- Hands-on experience with Vector CANoe, including IG block configuration, trace analysis, graphical panels, measurement setup, and signal monitoring.
- Collaborated with cross-functional teams to ensure feature compliance, quality validation, and on-time delivery.

SKILLS:

- **Testing:** Detection Testing, HMI Testing, Calibration, Diagnostics, Failsafe, Load Testing.
- **Methodologies:** STLC, SDLC, V-Model.
- **Protocols:** CAN, LIN, UART, SPI, I2C, UDS
- **Languages:** C, CAPL, Python
- **Validation Methods:** HIL Testing, Regression & Functional Testing, Manual & Automation Testing.
- **Tools & Platforms:** CANoe, ADTF, MITE, Automation Desk, Windchill (ALM), pytest, Jenkins, Git(Basics).

CERTIFICATIONS:

- ISTQB® Foundation Level (CTFL) – Udemy
- The Python Bible™ – Udemy
- Bash Shell Scripting – Udemy

ACHIEVEMENTS:

- Reduced project cost by automating test cases using Python, eliminating the dependency on dual automation licenses.
- Reduced testing cycle time by 20% through optimized test procedures and workflow improvements.

EDUCATION:

B.Tech – Electronics & Communication Engineering | 2019 – 2023
KKR & KSR Institute of Technology & Sciences

- Secured First Class with Distinction
- Final Year Project: Smart Parking System – Designed and implemented a prototype to monitor parking slot availability, log vehicle entry/exit times, and generate bills based on parking duration using embedded system logic.

LANGUAGES

- English
- Marwadi
- Hindi
- Telugu

DECLARATION

I hereby declare that the above-mentioned particulars are true to the best of my knowledge.

Naresh Kothari